



DO-IT-YOURSELF: PROJECT

PARTS LIST

Semiconductors:

Board1 - Arduino Nano board

Miscellaneous:

Module1 - LCD I2C (RG1602A)
- IR module
- 5V relay module
- Jumper wires
- Breadboard or general-purpose PCB

to the highest voltage source. Each of the 14 digital pins on the Nano board can be used as an input or output, using pinMode(), digitalWrite(), and digitalRead() functions. They operate at 5 volts. Each pin can provide or receive a maximum of 40mA current and has an internal pull-up resistor.

A typical I2C LCD (module1) display consists of an HD44780-based character LCD display and an I2C LCD adaptor. The LCD is ideal for displaying only characters. A 16x2 character LCD, for example, can display 32 ASCII characters across two rows. At the heart of the adaptor is an 8-bit I/O expander chip—PCF8574. This chip converts the I2C data from Arduino into the parallel data required for the LCD display.

The circuit's working is simple. During short-duration rest breaks or tea breaks, workers can turn on the fan, AC, or cooler just by waving their hand or standing in front of the sensor, without touching any switch or any loose connections. This system is compact and can be connected to any AC load through

a relay. This module can be used in various ways in different situations. Whenever a person comes in front of the sensor, the sensor will detect the presence of a human and turn on the connected load through the relay.

Software

The circuit is programmed using Arduino 1.8.19 tool. The IR sensor's analogue output signal is taken on pin D9 of Arduino Nano and it displays the status of the load (whether on/off) and the presence of a person on the LCD in digital form. The relay is connected to pin D12, through which we can connect loads like fans, coolers, and AC units. The LCD shows the status of the load and presence/absence of a person.

The source code is written in Arduino and saved as Loadcontrol.ino. Before uploading the sketch program to the Arduino Nano board, do not forget to select the proper port. After selecting the proper port, upload the software file Loadcontrol.ino to the board.

Construction and testing

After uploading the source code Loadcontrol.ino to Arduino Nano, assemble the circuit on a breadboard. Alternatively, you can assemble it on a general-purpose PCB. After wiring the circuit, enclose it in a suitable box, including the Arduino Nano board.

Check connections as per the circuit diagram. Provide voltage to the assembled circuit from the Arduino Nano board. Check the status of the load shown on the LCD. Always advise the workers to move in front of the sensor to activate it. **EFY**

EFY Note

The source code of this project is available for free download at www.electronicsforu.com



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